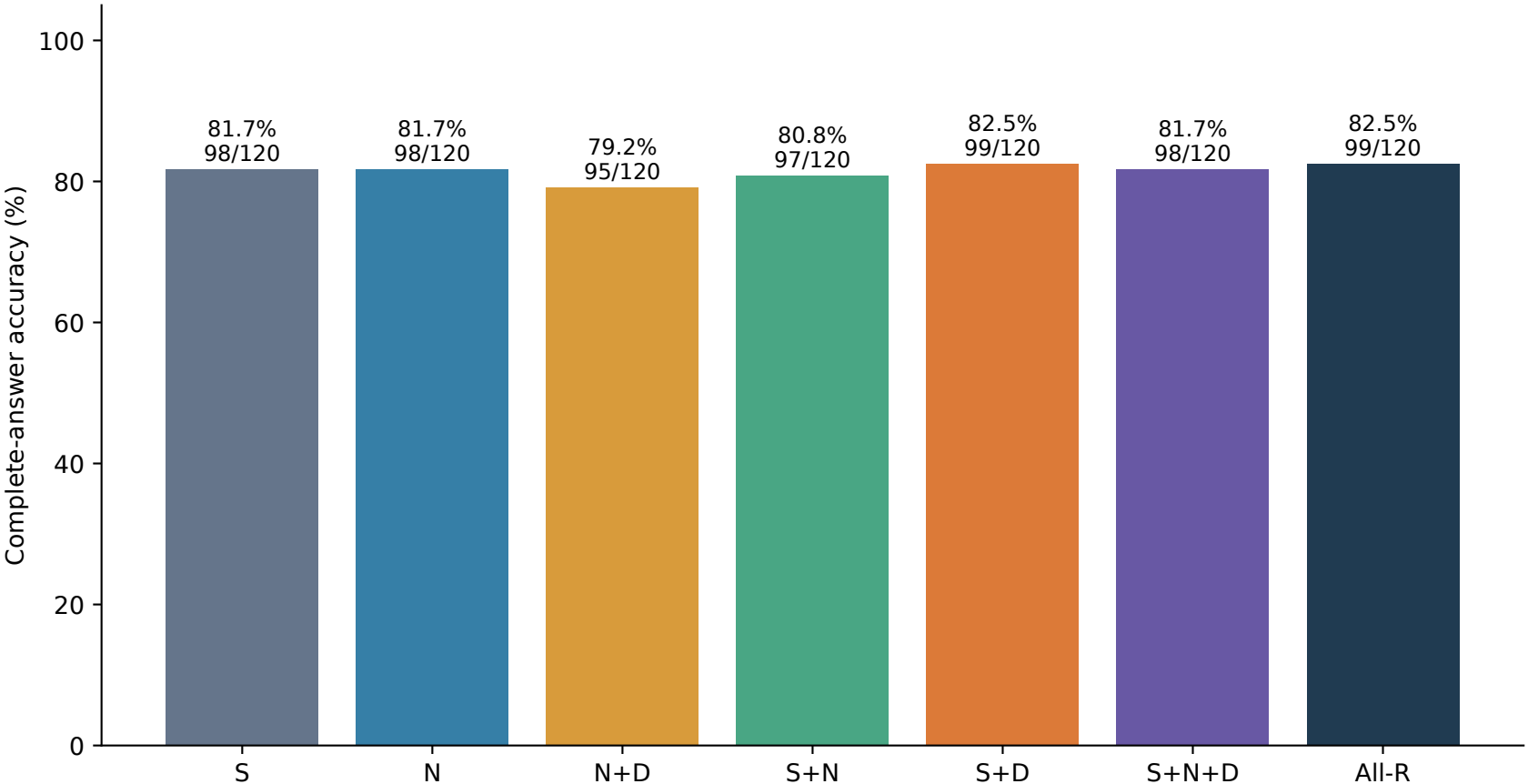
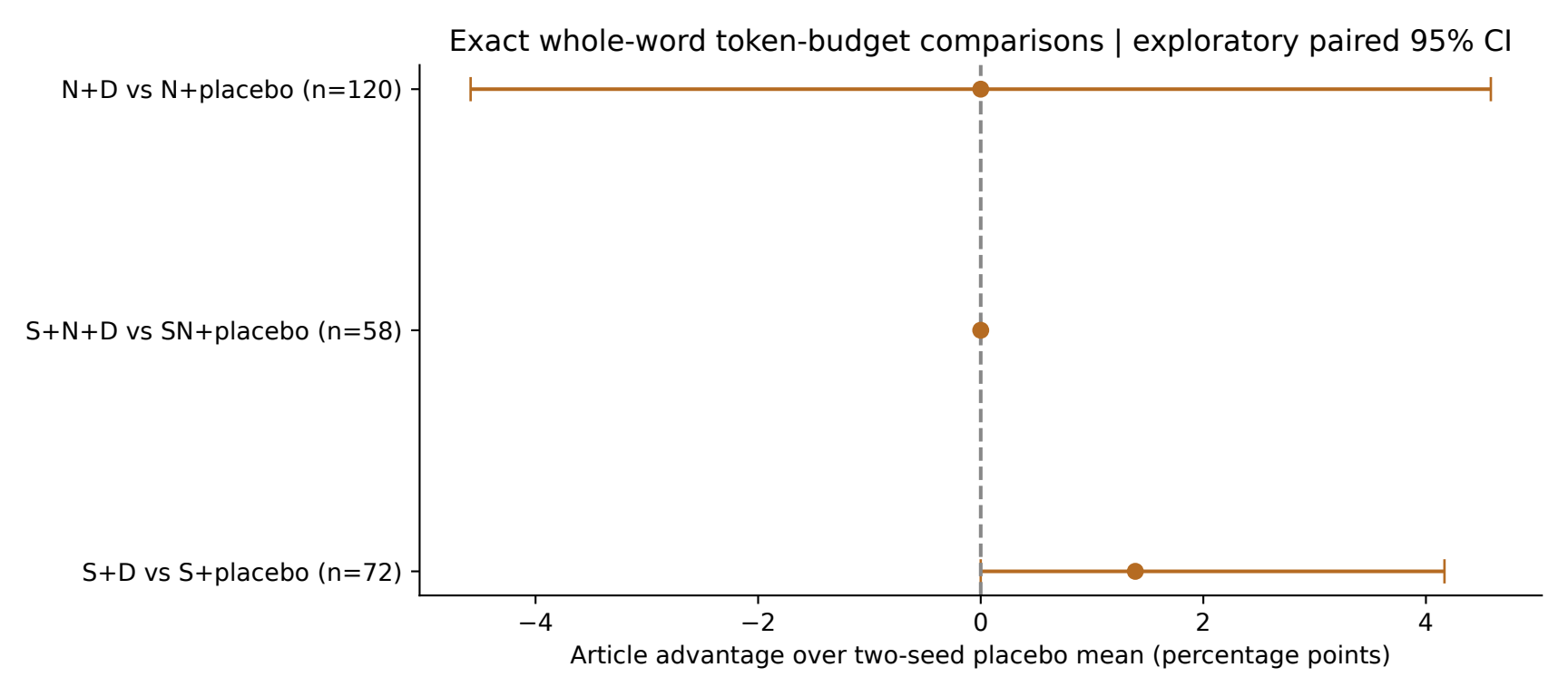
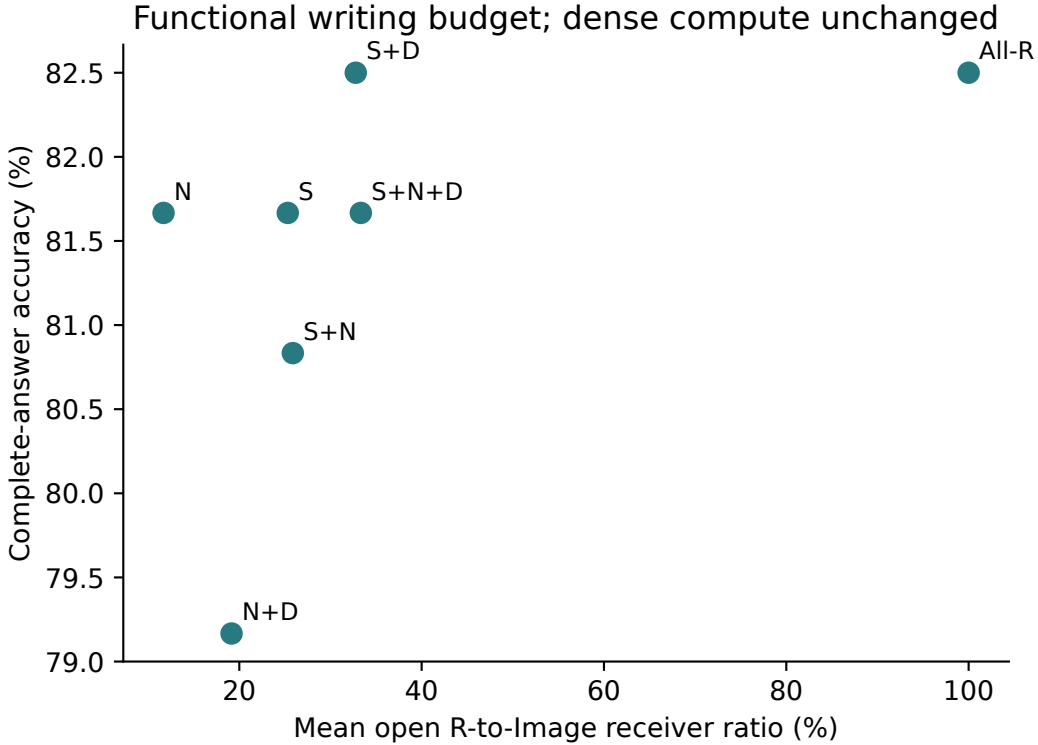
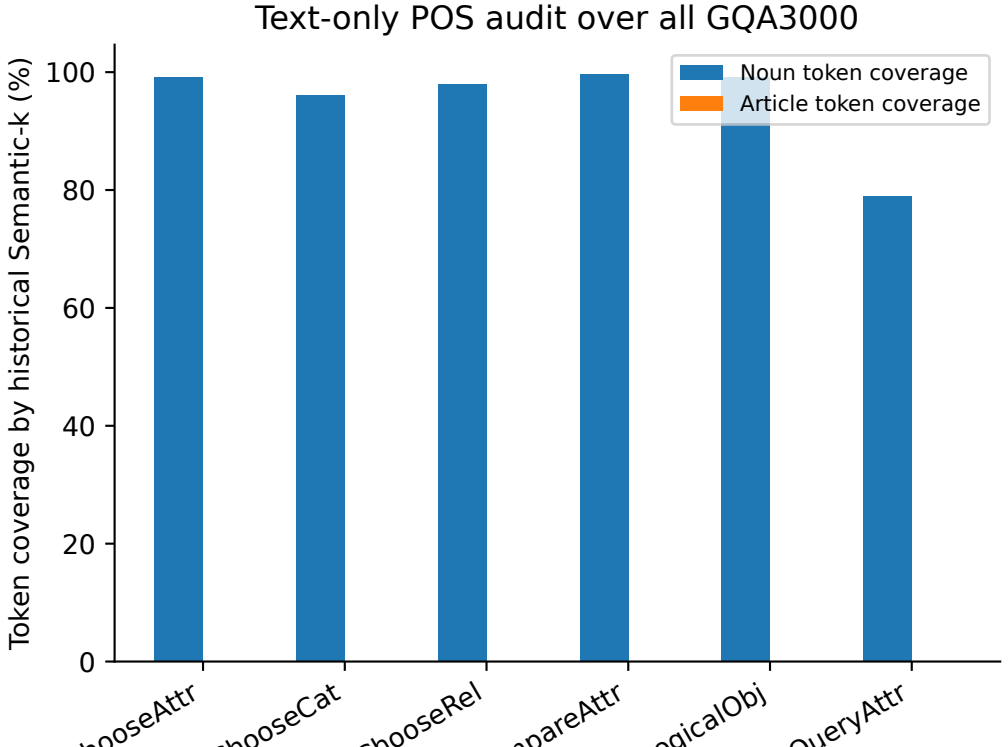

Noun + Article receiver interventions | 120 image-disjoint questions
Complete-answer accuracy (%)
0
20
40
60
80
100
81.7%
98/120
81.7%
98/120
79.2%
95/120
80.8%
97/120
82.5%
99/120
81.7%
98/120
82.5%
99/120
S
N
N+D
S+N
S+D
S+N+D
All-R


Exact whole-word token-budget comparisons | exploratory paired 95% CI
N+D vs N+placebo (n=120)
S+N+D vs SN+placebo (n=58)
S+D vs S+placebo (n=72)
−4
−2
0
2
4
Article advantage over two-seed placebo mean (percentage points)


Functional writing budget; dense compute unchanged
Complete-answer accuracy (%)
Mean open R-to-Image receiver ratio (%)
82.5
82.0
81.5
81.0
80.5
80.0
79.5
79.0
20
40
60
80
100
S+D
All-R
N
S
S+N+D
S+N
N+D


Text-only POS audit over all GQA3000
Token coverage by historical Semantic-k (%)
Noun token coverage
Article token coverage
100
80
60
40
20
0
ChooseAttr
ChooseCat
ChooseRel
CompareAttr
LogicalObj
QueryAttr